Vehicle Management System

A computerized management system that helps the barangay organize, monitor and manage its emergency response vehicles.

MUST HAVES

- A web-based platform that works on desktop and mobile browsers.
- Role-based access for Super Admin, Admin, Custodian and Maintenance Personnel; each account has one role.
- Account registration with a barangay staff code, a valid ID and a photo, followed by approval before the account can sign in. Accounts an Admin creates start with a temporary password that the user must replace at first sign-in.
- Organized, up-to-date vehicle records, vehicle types and current locations.
- A maintenance workflow: issue report (Custodian), ticket proposal, Admin approval, mechanic assignment, repair, and Custodian verification before closing.
- Repairs that can use parts from one or more donor vehicles, or be sent to an external shop; condition updates automatically as tickets are raised and closed.
- Maintenance records and recurring schedules, covering in-house repairs and work sent to an external shop.
- Dashboard summaries for each role.
- Cloud storage for uploaded images and files (issue photos, repair attachments and vehicle documents), with the data in a MySQL database.
- Email notices for registration approval and staff codes, plus in-app notifications.

Admin functionalities

- Manage vehicle records, vehicle types and locations.
- Monitor vehicle conditions and readiness to respond.
- View issue reports and approve, decline or request corrections on ticket proposals (including external-shop proposals, which the Admin only approves).
- Assign Maintenance Personnel and manage maintenance records and schedules.
- View each vehicle's history.
- Generate printable and CSV reports (vehicles, issues, maintenance, schedules, history).
- Monitor the activity log for accountability.
- Approve staff registrations for their barangay and create user accounts with a temporary password.

Custodian functionalities

- View, search and filter vehicle information and availability.
- Report vehicle issues with photo evidence and track their status (only the Custodian reports issues).

- Check vehicle condition and propose maintenance tickets, linking them to an open issue report when one applies.
- View vehicle types and add a type; view locations and add or edit a location.
- Log work sent to an external shop (send out, receive back, rework), then test the vehicle and close the ticket by verifying; the Admin only approves it.
- Verify completed repairs before a ticket is closed.
- Book maintenance schedules and view maintenance status.

Maintenance Personnel functionalities

- View assigned work orders and every vehicle in the barangay.
- Log repair actions, parts used (including parts taken from donor vehicles), evidence files and remarks.
- Update repair progress and completion.
- Complete assigned maintenance schedules and monitor upcoming maintenance.
- View maintenance history.

Super Admin functionalities (platform level)

- Review and approve or reject registrations, checking the applicant's ID and photo.
- Manage barangays, users and registration codes, and email staff codes.
- View the platform activity log and handle Support Center concerns.